For Loops in Python

Known Repetition, Collections and Nested Loops

Programming Fundamentals Team

SETU

2026-06-03

Outline

Outline

1.1 By the end of this lecture

You should be able to:

- Explain when a `for` loop is useful
- Use `range()` with start, stop and step values
- Iterate through strings
- Iterate through lists
- Use selection inside a loop
- Use accumulators with `for` loops
- Write nested loops
- Create simple text patterns

Outline

2.1 Review: while loop

```
i = 1

while i <= 10:
    print(i)
    i += 1
```

Use `while` when we may not know how many repetitions are needed.

2.2 A simpler way when we know the count

```
for i in range(1, 11):
    print(i)
```

A `for` loop is useful when we know what sequence we want to process.

2.3 While versus for

while loop

Use when repetition depends on a condition.

Examples:

- password validation
- menu loops
- sentinel loops

for loop

Use when processing a sequence.

Examples:

- numbers in a range
- characters in a string
- items in a list

Outline

3.1 Syntax

```
for variable in sequence:
    statements
```

Examples of sequences:

- `range(5)`
- a string such as `"Python"`
- a list such as `[70, 80, 90]`

3.2 First for loop

```
for i in range(5):
    print(i)
```

Output:

```
0
1
2
3
4
```

3.3 Important: range stops before the end

```
range(5)
```

Produces:

```
0, 1, 2, 3, 4
```

It does not include 5.

Outline

4.1 Start and stop

```
for i in range(1, 11):
    print(i)
```

This prints 1 to 10.

`range(start, stop)` includes the start but excludes the stop.

4.2 Step value

```
for i in range(0, 21, 2):
    print(i)
```

This prints even numbers from 0 to 20.

4.3 Counting backwards

```
for i in range(10, 0, -1):
    print(i)
```

This prints 10 down to 1.

4.4 Activity: predict the output

```
for i in range(2, 12, 3):
    print(i)
```

Questions:

- What is the first value?
- What is added each time?
- Why is 12 not printed?

4.5 Quick challenges

Write `for` loops to print:

1. 1 to 20
2. 10 to 1
3. Multiples of 5 from 5 to 50
4. Odd numbers from 1 to 19

Outline

5.1 Sum numbers 1 to 100

```
total = 0

for i in range(1, 101):
    total += i

print(total)
```

5.2 Calculate an average

```
total = 0

for i in range(5):
    mark = int(input("Enter mark: "))
    total += mark

average = total / 5
print("Average:", average)
```

5.3 Add validation

```
total = 0

for i in range(5):
    mark = int(input("Enter mark: "))

    while mark < 0 or mark > 100:
        mark = int(input("Invalid. Try again: "))

    total += mark

print("Average:", total / 5)
```

This combines `for`, `while`, and `or`.

Outline

6.1 Strings are sequences

```
for letter in "Python":
    print(letter)
```

Output:

```
P
y
t
h
o
n
```

6.2 Count vowels

```
word = input("Enter word: ")
vowels = 0

for letter in word:
    if letter in "aeiouAEIOU":
        vowels += 1

print("Vowels:", vowels)
```

6.3 Activity: count spaces

Write a program that asks for a sentence and counts how many spaces it contains.

Hint:

```
if character == " ":
```

6.4 Solution: count spaces

```
sentence = input("Enter a sentence: ")

space_count = 0

for character in sentence:
    if character == " ":
        space_count += 1

print("Number of spaces:", space_count)
```

Outline

7.1 What is a list?

A list stores multiple values in one variable.

```
marks = [75, 60, 80, 90]
```

A `for` loop is ideal for processing every item in a list.

7.2 Print each mark

```
marks = [75, 60, 80, 90]

for mark in marks:
    print(mark)
```

7.3 Total and average

```
marks = [75, 60, 80, 90]
total = 0

for mark in marks:
    total += mark

average = total / len(marks)
print("Average:", average)
```

7.4 Find the highest mark

```
marks = [75, 60, 80, 90]
highest = marks[0]

for mark in marks:
    if mark > highest:
        highest = mark

print("Highest:", highest)
```

7.5 Count passes

```
marks = [75, 40, 32, 81, 55]
passes = 0

for mark in marks:
    if mark >= 40:
        passes += 1

print("Passes:", passes)
```

Outline

8.1 What is a nested loop?

A nested loop is a loop inside another loop.

Outer loop controls the main repetition.

Inner loop completes all of its repetitions for each one repetition of the outer loop.

8.2 Rows and Seats Analogy

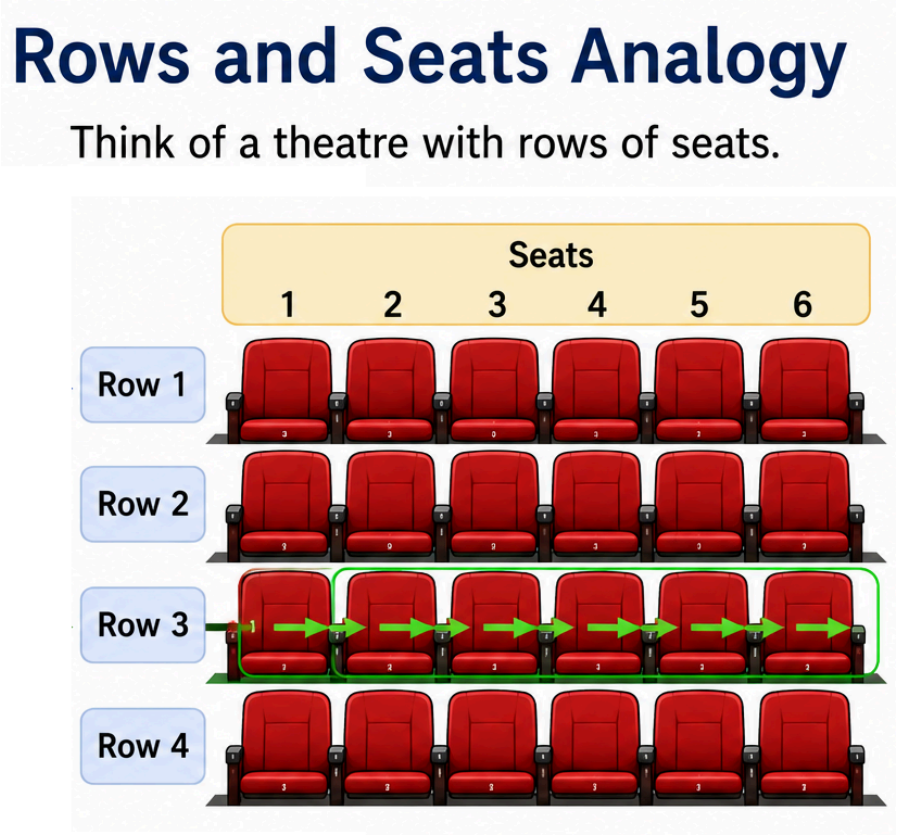
Imagine a classroom:

- 3 rows
- 4 seats in each row

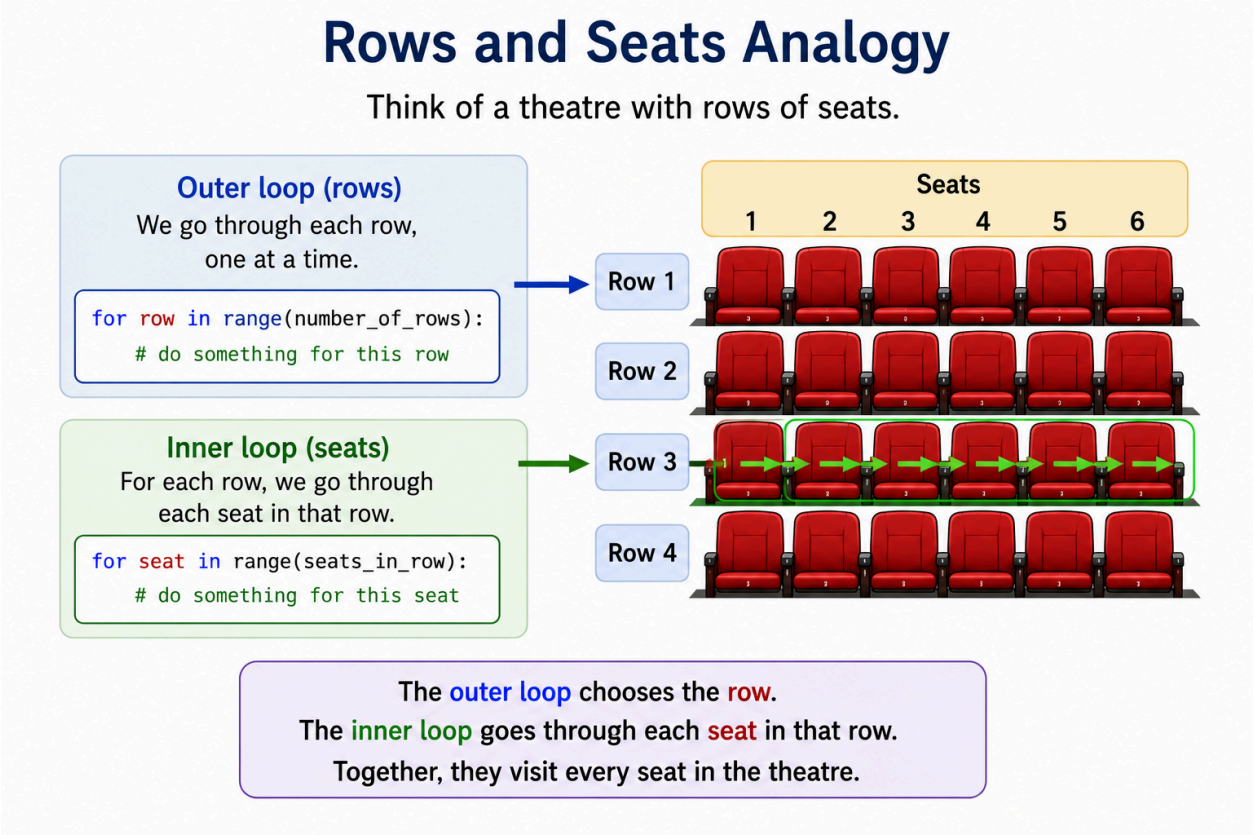
For every row, check every seat.

The **outer loop** selects the row.

The **inner loop** selects each seat in that row.



8.3 Rows and Seats Analogy Continued



8.4 Rows and Seats Analogy Code

```
for row in range(4):
    for seat in range(6):
        print(row, seat)
```

8.5 Trace the idea

```
row 0: seat 0, seat 1, seat 2, seat 3, seat 4, seat 5
row 1: seat 0, seat 1, seat 2, seat 3, seat 4, seat 5
row 2: seat 0, seat 1, seat 2, seat 3, seat 4, seat 5
row 3: seat 0, seat 1, seat 2, seat 3, seat 4, seat 6
```

The inner loop runs completely for each value of `row`.

8.6 Is this code better?

```
for row in range(1,5):
    print("\nRow ",row, end=' ')
    for seat in range(1,7):
        print(" Seat ", seat, end=' ')
```

Outline

9.1 Print a square

```
for row in range(5):
    print("*" * 5)
```

Output:

```
*****
*****
*****
*****
*****
```

9.2 Print a triangle

```
for row in range(1, 6):
    print("*" * row)
```

Output:

```
*
**
***
****
*****
```

9.3 Multiplication table

```
for i in range(1, 11):
    print(i, "x 7 =", i * 7)
```

9.4 1 to 10 tables

```
for table in range(1, 11):
    print("Table", table)

    for i in range(1, 11):
        print(i, "x", table, "=", i * table)

    print()
```

Outline

10.1 Off-by-one errors

```
for i in range(1, 10):
    print(i)
```

This prints 1 to 9, not 1 to 10.

The stop value is excluded.

10.2 Indentation errors

```
for i in range(5):
print(i)
```

Python needs indentation to show what belongs inside the loop.

10.3 Changing the wrong variable

```
total = 0

for mark in [50, 60, 70]:
    total = mark
```

This replaces the total each time.

Better:

```
total += mark
```

Outline

11.1 Mini-program: mark report

```
marks = [72, 38, 55, 90, 41]

total = 0
passes = 0
highest = marks[0]

for mark in marks:
    total += mark

    if mark >= 40:
        passes += 1

    if mark > highest:
        highest = mark
```

11.1 Mini-program: mark report

```
print("Average:", total / len(marks))
print("Passes:", passes)
print("Highest:", highest)
```

11.2 What does this combine?

- Lists
- `for` loops
- Selection
- Counters
- Accumulators
- Highest value pattern
- `len()` function

Outline

12.1 Today we learned

- Use `for` loops to process a sequence.
- `range()` generates numbers.
- The stop value in `range()` is excluded.
- Strings and lists can be looped through directly.
- Selection can be placed inside loops.
- Nested loops are useful for grids, tables and patterns.

12.2 Practice exercises

1. Print numbers from 1 to 50.
2. Print multiples of 3 from 3 to 60.
3. Print each character in your name.
4. Count vowels in a sentence.
5. Calculate the total of a list of numbers.
6. Count how many marks are passes.
7. Find the highest number in a list.
8. Print a 5 by 5 square of stars.
9. Print a right-angled triangle.
10. Challenge: print multiplication tables from 1 to 10.